

Week 5 Assignment Report

Objective

The objective of this assignment is to develop practical skills in Power BI and Business Intelligence. Students will learn to import and prepare datasets, create data models, build calculated measures using DAX, create interactive visualizations, and develop an interactive dashboard for business analysis.

Task 1: Power BI Introduction & Data Import

Using Power BI: Explore the Power BI interface. Import a suitable dataset. View the imported data. Identify the available tables and columns. Check the data types of the columns. Explain the basic purpose of Power BI in Data Analytics.

Dataset Used

A Sales dataset was imported into Power BI Desktop using an Excel workbook. The workbook contains three tables:

1. **Sales** – Contains order details, dates, customer IDs, product IDs, quantities, and sales amounts.
2. **Customers** – Contains customer information such as customer ID, customer name, city, and region.
3. **Products** – Contains product information such as product ID, product name, category, and unit price.

Data Import

The Excel workbook was imported into Power BI Desktop using the **Home → Excel Workbook** option. The Sales, Customers, and Products tables were selected and loaded into Power BI.

Available Tables and Columns

Sales Table:

- Order ID
- Date
- Customer ID
- Product ID
- Quantity
- Sales Amount

Customers Table:

- Customer ID
- Customer Name
- City
- Region

Products Table:

- Product ID
- Product Name
- Category
- Unit Price

Data Types

The numerical columns such as Order ID, Customer ID, Product ID, Quantity, and Sales Amount were identified as numeric data types. Date was identified as a Date data type, while names, cities, regions, product names, and categories were identified as Text data types.

Basic Purpose of Power BI in Data Analytics

Power BI is a Business Intelligence and data visualization tool used to connect to different data sources, prepare and transform data, analyze information, create interactive visualizations, and develop dashboards. It helps organizations understand trends and patterns in their data and supports data-driven decision-making.

Conclusion

The dataset was successfully imported into Power BI Desktop. The available tables, columns, and data types were examined, providing a foundation for data cleaning, modeling, DAX calculations, and visualization in the following tasks.

-----output-----

Navigator

Display Options ▾

PowerBI_Sales_Assignment_Dataset.xlsx [3]

☒ Customers

☒ Products

☒ Sales

Sales

Order ID	Date	Customer ID	Product ID	Quantity	Sales Amount
1001	46027	1	101	2	
1002	46030	2	102	1	
1003	46037	3	103	3	
1004	46044	4	104	1	
1005	46056	5	105	2	
1006	46063	1	106	1	
1007	46071	6	107	4	
1008	46078	7	108	2	
1009	46085	8	109	1	
1010	46093	9	110	3	
1011	46101	10	101	1	
1012	46109	2	103	2	
1013	46118	3	104	2	
1014	46126	4	105	1	
1015	46133	5	108	3	
1016	46144	6	109	2	
1017	46152	7	110	2	
1018	46160	8	102	3	
1019	46174	9	106	2	
1020	46188	10	107	3	
1021	46198	1	103	1	
1022	46207	2	104	1	
1023	46215	3	108	2	

Load

Transform Data

Cancel

Load

- Products
Schema synchronized
- Customers
Schema synchronized
- Sales
Schema synchronized

Cancel

FileHomeHelpTable toolsColumn tools

NameCustomer ID

FormatWhole number

SummarizationDon't summarize

Sort by columnSort

Data typeWhole number

Format\$ - % 0

Data categoryUncategorized

Data groupsData groups

Manage relationshipsManage relationships

New columnNew column

StructureFormattingProperties

Customer IDCustomer NameCityRegion

1Aarav ShahAhmedabadWest

2Riya PatelSuratWest

3Rahul MehtaMumbaiWest

4Priya DesaiPuneWest

5Karan JoshiJaipurNorth

6Neha SinghDelhiNorth

7Arjun VermaLucknowNorth

8Isha KapoorBengaluruSouth

9Vivaan RaoHyderabadSouth

10Ananya IyerChennaiSouth

Search

CustomersCityCustomer IDCustomer NameRegionProductsCategoryProduct IDProduct NameUnit PriceSalesCustomer IDDateOrder IDProduct IDQuantitySales Amount

Table: Customers (10 rows) Column: Customer ID (10 distinct values)

FileHomeHelpTable toolsColumn tools

NameCategory

FormatText

SummarizationDon't summarize

Sort by columnSort

Data typeText

Format\$ - % Auto

Data categoryUncategorized

Data groupsData groups

Manage relationshipsManage relationships

New columnNew column

StructureFormattingProperties

Product IDProduct NameCategoryUnit Price

101LaptopElectronics1200

102SmartphoneElectronics1800

103HeadphonesAccessories1500

104MonitorElectronics3200

105KeyboardAccessories1300

106MouseAccessories1500

107PrinterOffice Equipment700

108TabletElectronics1800

109SmartwatchWearables2200

110SpeakerAudio1300

File Home Help Table tools Column tools					
Name	Quantity	Format	Whole number	Summarization	Sum
Data type	Whole number	Format	\$ % 0	Data category	Uncategorized
Structure	Formatting	Properties	Sort by column	Data groups	Manage relationships
Sort	Groups	Relationships	New column	Calculations	

Order ID	Date	Customer ID	Product ID	Quantity	Sales Amount
1001	46027	1	101	2	2400
1002	46030	2	102	1	1800
1003	46037	3	103	3	4500
1004	46044	4	104	1	3200
1005	46056	5	105	2	2600
1006	46063	1	106	1	1500
1007	46071	6	107	4	2800
1008	46078	7	108	2	3600
1009	46085	8	109	1	2200
1010	46093	9	110	3	3900
1011	46101	10	101	1	1200
1012	46109	2	103	2	3000
1013	46118	3	104	2	6400
1014	46126	4	105	1	1300
1015	46133	5	108	3	5400
1016	46144	6	109	2	4400
1017	46152	7	110	2	2600
1018	46160	8	102	3	5400
1019	46174	9	106	2	3000
1020	46188	10	107	3	2100
1021	46198	1	103	1	1500
1022	46207	2	104	1	3200
1023	46215	3	108	2	3600
1024	46223	4	110	3	3900
1025	46239	5	101	2	2400
1026	46248	6	105	2	2600
1027	46256	7	107	5	3500
1028	46264	8	109	2	4400

Table: Sales (28 rows) Column: Quantity (5 distinct values)

File Home Help					
Clipboard	Get data	Excel	OneLake	SQL	Enter data
Recent sources	Transform data	Refresh	Manage relationships	New measure	New table
Calculations	New calculation group	New parameter	Manage roles	View as	Q&A
Language	Linguistic	Schema	Sensitivity	Publish	Share

Products	Sales	Customers
Category	Customer ID	City
Product ID	Date	Customer ID
Product Name	Order ID	Customer Name
Unit Price	Product ID	Region
	Quantity	
	Sales Amount	

Name	Customers
Description	Enter a description
Row label	Select a row label
Is hidden	No

DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#)

Run Update model with changes (0)

```

1 // Welcome to DAX query view! Learn more about DAX queries at https://aka.ms/dax-queries.
2 // Right-click on tables, columns, or measures in the data pane to access quick queries, or ask Copilot for help writing DAX.
3

```

↑ Apply

Preview

- 1 Welcome to TMDL View! Learn more about TMDL at <https://aka.ms/tmdl-view>

To start editing, drag any semantic model object from the Data pane, or script with TMDL from scratch.
When you're ready to apply your changes, click "Apply."

Here's an example of TMDL Script:

```
createOrReplace
```

```
    table Calendar
```

```
        column Date
```

```
            dataType: dateTime
```

```
            isKey
```

```
            formatString: yyyy-mm-dd
```

```
            summarizeBy: none
```

```
            sourceColumn: Date
```

Task 2: Data Cleaning & Preparation

Using Power Query: Inspect the imported dataset. Identify missing or incorrect values. Remove duplicate records where required. Change incorrect data types. Rename columns where appropriate. Remove unnecessary columns. Apply suitable transformations to prepare the dataset for analysis.

Power Query Editor in Power BI was used to inspect and prepare the imported dataset for analysis.

Data Inspection

The Sales, Customers, and Products tables were inspected for missing values, duplicate records, incorrect data types, and unnecessary columns.

Missing Values

The dataset was checked for null or blank values. No significant missing values were identified in the tables.

Duplicate Records

The Sales table was checked for duplicate records using the Remove Duplicates option in Power Query. The records were unique, so no important records were removed.

Data Type Correction

The data types were reviewed and set appropriately. ID and quantity fields were assigned the Whole Number data type, the Date field was assigned the Date data type, and descriptive fields such as names, cities, regions, and categories were assigned the Text data type.

Column Renaming

The Sales Amount column in the Sales table was renamed to Total Sales to make the field more descriptive and easier to understand during analysis.

Unnecessary Columns

All available columns were relevant to the business analysis, so no unnecessary columns were removed.

Data Preparation

The dataset was inspected and transformed using Power Query. After completing the required preparation steps, the changes were applied using Close & Apply.

Conclusion

Power Query was successfully used to inspect, clean, validate, and prepare the dataset for data modeling and further analysis in Power BI.

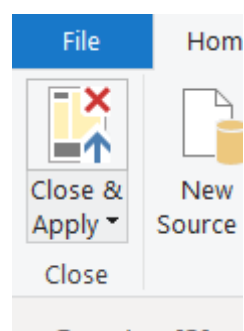
-----output-----

5 COLUMNS, 28 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 08:39 A

	123 Sales Amount	
2	2400	
1	1800	
3	4500	
1	3200	
2	2600	
1	1500	
4	2800	
2	3600	
1	2200	
3	3900	
1	1200	
2	3000	
2	6400	
1	1300	
3	5400	
2	4400	
2	2600	
3	5400	

	123 Total sales
2	2400
1	1800
3	4500
1	3200
2	2600
1	1500
4	2800
2	3600
1	2200
3	3900
1	1200
2	3000
2	6400
1	1300
3	5400
2	4400
2	2600
3	5400
2	3000
3	2100
1	1500



Task 3: Data Modeling & Relationships

Using a dataset containing multiple tables: Identify common columns between tables. Create relationships between relevant tables. Select the appropriate relationship type. Create a basic data model. Explain the relationships created.

The dataset contains three related tables: Sales, Customers, and Products. The common columns between these tables were identified and used to create relationships in Power BI.

Relationships Created

1. Customers – Sales

The Customer ID column was used to establish a relationship between the Customers and Sales tables.

- Customers[Customer ID] → Sales[Customer ID]
- Relationship: **One-to-Many (1:*)**

One customer can have multiple sales records, while each sales record is associated with one customer.

2. Products – Sales

The Product ID column was used to establish a relationship between the Products and Sales tables.

- Products[Product ID] → Sales[Product ID]
- Relationship: **One-to-Many (1:*)**

One product can appear in multiple sales records, while each sales record is associated with one product.

Basic Data Model

The Sales table acts as the central transaction table, while Customers and Products provide additional descriptive information. This model allows sales to be analyzed by customer, city, region, product, and category.

Conclusion

The relationships between the three tables were successfully created using common ID columns. A one-to-many relationship was appropriate because one customer or product can be associated with multiple sales transactions.

New relationship



Select tables and columns that are related.

From table

Sales

▼

Customer ID	Date	Order ID	Product ID	Quantity	Total sales
1	46027	1001	101	2	2400
2	46030	1002	102	1	1800
3	46037	1003	103	3	4500

To table

Customers

▼

City	Customer ID	Customer Na...	Region
Ahmedabad	1	Aarav Shah	West
Surat	2	Riya Patel	West
Mumbai	3	Rahul Mehta	West

Cardinality

Many to one (*:1)

▼

Cross-filter direction

Single

▼

☒ Make this relationship active

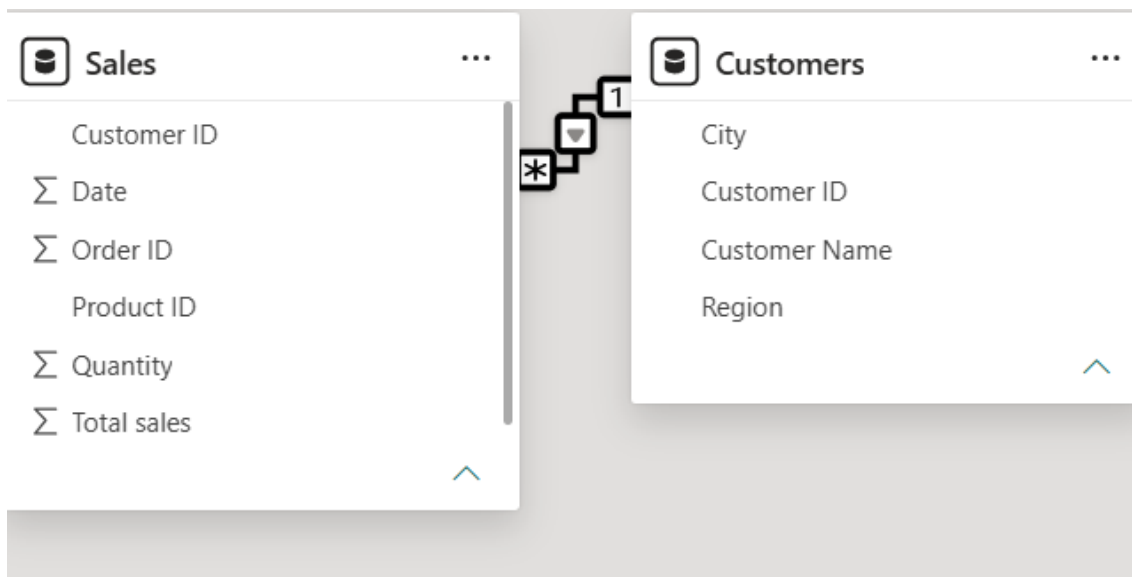
☐ Apply security filter in both directions

☐ Assume referential integrity

✖ There's already a relationship between these two columns.

Save

Cancel



New relationship

Select tables and columns that are related.

From table

Sales

Customer ID	Date	Order ID	Product ID	Quantity	Total sales
1	46027	1001	101	2	2400
2	46030	1002	102	1	1800
3	46037	1003	103	3	4500

To table

Products

Category	Product ID	Product Name	Unit Price
Electronics	101	Laptop	1200
Electronics	102	Smartphone	1800
Accessories	103	Headphones	1500

Cardinality

Many to one (*:1)

☒ Make this relationship active

☐ Assume referential integrity

Cross-filter direction

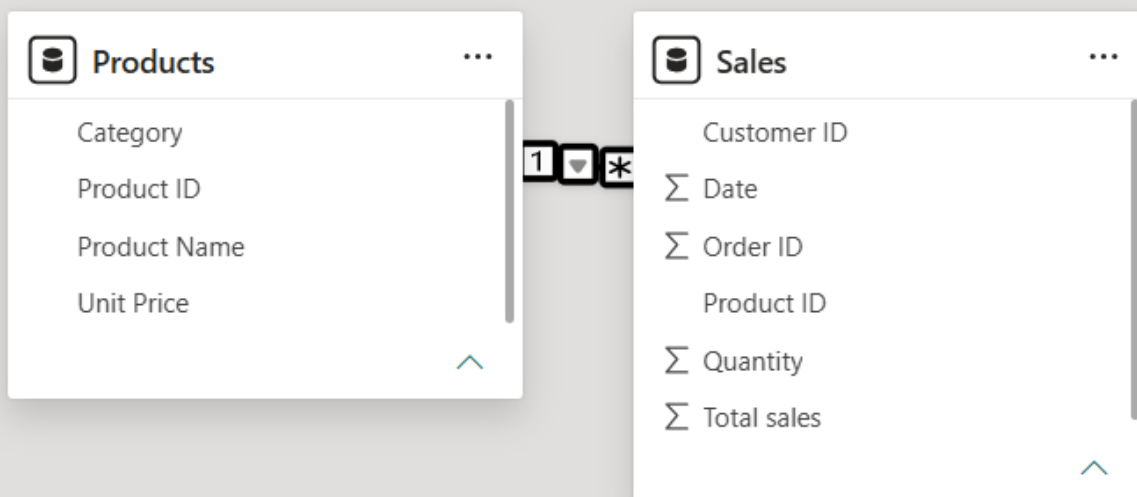
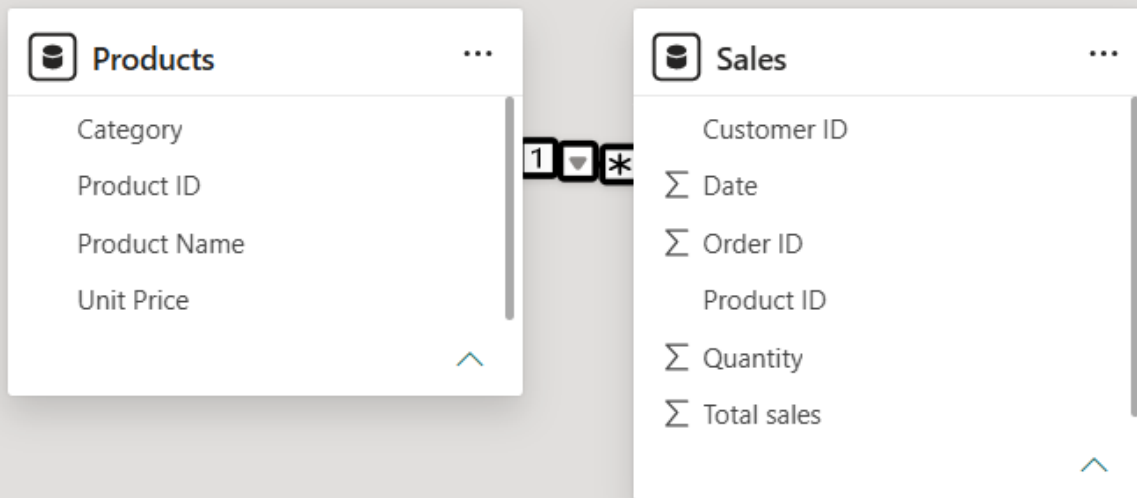
Single

☐ Apply security filter in both directions

✖ There's already a relationship between these two columns.

Save

Cancel



Task 4: DAX Calculated Columns & Measures

Create DAX expressions to perform calculations such as: Total Sales / Total Amount. Average Sales / Average Value. Total Number of Records. Maximum and Minimum Values. Percentage or other relevant business calculations. Create at least 3 DAX measures and display their results.

Task 4: DAX Calculated Columns & Measures

DAX Measures Created

DAX (Data Analysis Expressions) was used in Power BI to create calculated measures for analyzing the Sales data. Measures perform calculations dynamically based on the data and filters applied to the report.

1. Total Sales Amount

A measure was created to calculate the total sales amount from the Sales table.

Total Sales = SUM(Sales[Sales Amount])

This measure calculates the sum of all sales amounts recorded in the dataset.

2. Average Sales Amount

A measure was created to calculate the average sales amount.

Average Sales = AVERAGE(Sales[Sales Amount])

This measure calculates the average value of sales transactions.

3. Total Number of Records

A measure was created to count the total number of records in the Sales table.

Total Records = COUNTROWS(Sales)

This measure displays the total number of sales records available in the dataset.

4. Maximum Sales Amount

A measure was created to identify the highest sales amount.

Maximum Sales = MAX(Sales[Sales Amount])

This measure returns the maximum sales amount recorded in the dataset.

5. Minimum Sales Amount

A measure was created to identify the lowest sales amount.

Minimum Sales = MIN(Sales[Sales Amount])

This measure returns the minimum sales amount recorded in the dataset.

Displaying the Results

The created DAX measures were displayed using **Card visuals** in Power BI. Separate Card visuals were used to display Total Sales, Average Sales, Total Records, Maximum Sales, and Minimum Sales.

Results

The report displays the calculated values dynamically through the DAX measures. These measures provide useful information about the overall sales performance, average transaction value, number of records, and sales range.

Conclusion

DAX measures were successfully created and used to perform calculations on the Sales dataset. The calculated results were displayed using Card visuals, providing a clear summary of important sales-related information.

-----output-----

Sum Sales 88K	Average Sales 3.16K	Total Records 28	Minimum Sales 1K	Maximum Sales 6K
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Task 5: Power BI Visualizations

Task 5: Power BI Visualizations

Visualizations Created

Power BI visualizations were created to analyze and present the Sales dataset in an interactive and meaningful manner. Different types of charts and visuals were used to represent sales trends, category-wise sales, and important summary information.

1. Bar/Column Chart – Sales by Product Category

A **Column Chart** was created to compare sales amounts across different product categories.

Fields used:

- **X-axis:** Category
- **Y-axis:** Sales Amount
- **Aggregation:** Sum

Title: Total Sales by Product Category

This visualization helps compare the sales performance of different product categories.

2. Line Chart – Sales Trend Over Time

A **Line Chart** was created to show how sales changed over time.

Fields used:

- **X-axis:** Date
- **Y-axis:** Sales Amount
- **Aggregation:** Sum

Title: Sales Trend Over Time

This visualization helps identify increases, decreases, and overall trends in sales over different dates.

3. Pie/Donut Chart – Sales by Category

A **Donut Chart** was created to represent the contribution of each product category to total sales.

Fields used:

- **Legend:** Category
- **Values:** Sales Amount
- **Aggregation:** Sum

Title: Sales Distribution by Category

This visualization shows the proportion of total sales contributed by each category.

4. Table – Sales Details

A **Table** visual was created to display important sales information in a structured format.

Fields used:

- Order ID
- Date
- Customer ID
- Product ID
- Quantity
- Sales Amount

Title: Sales Transaction Details

This table provides a detailed view of individual sales transactions.

5. KPI/Card – Total Sales

A **Card** visual was created to display the overall sales amount using the DAX measure created in Task 4.

Field used:

- Total Sales

Title: Total Sales Amount

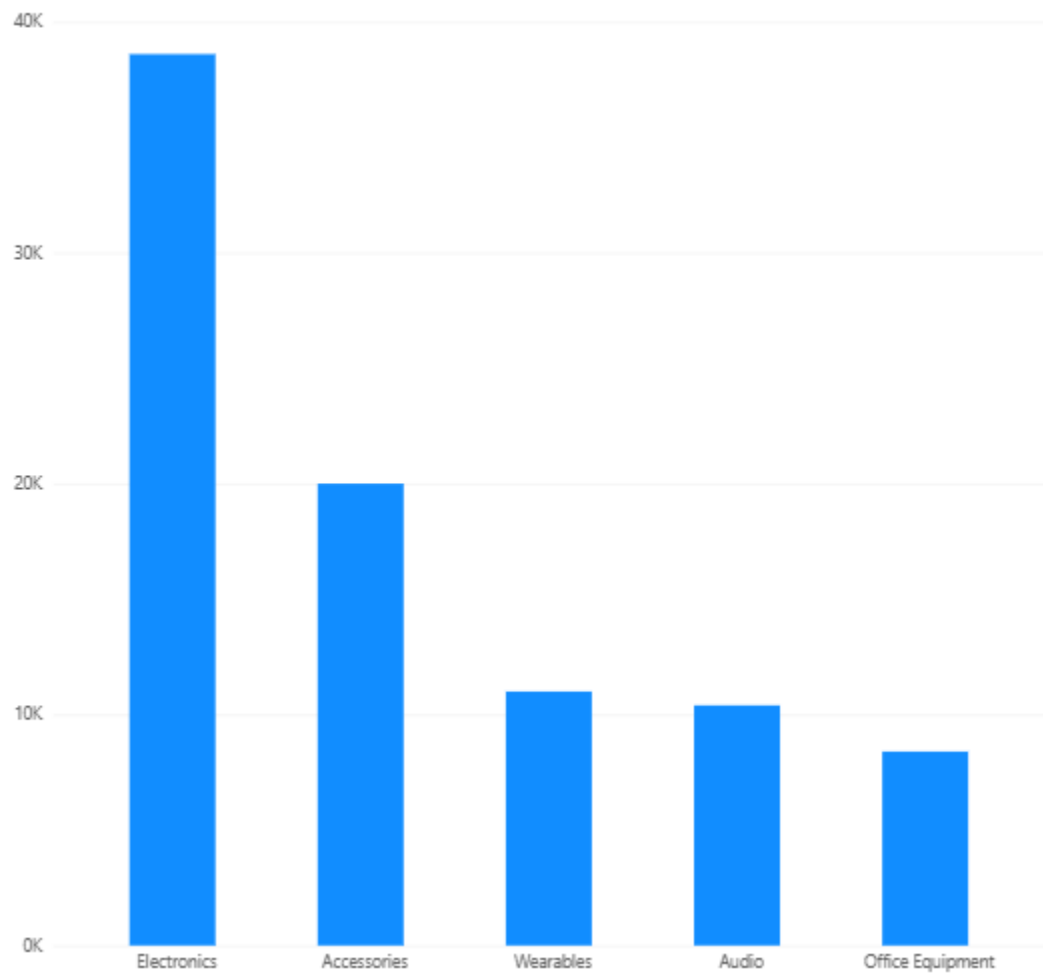
The Card provides a quick summary of the total sales value.

Conclusion

Multiple Power BI visualizations were successfully created using the Sales, Customers, and Products data. Bar/Column Charts were used for category comparisons, Line Charts for time-based trends, Donut Charts for sales distribution, Tables for detailed records, and Cards for displaying key performance indicators. These visualizations make the sales data easier to understand and support effective data analysis and decision-making.

Total Sales by Product Category

Sum of Sales Amount by Category

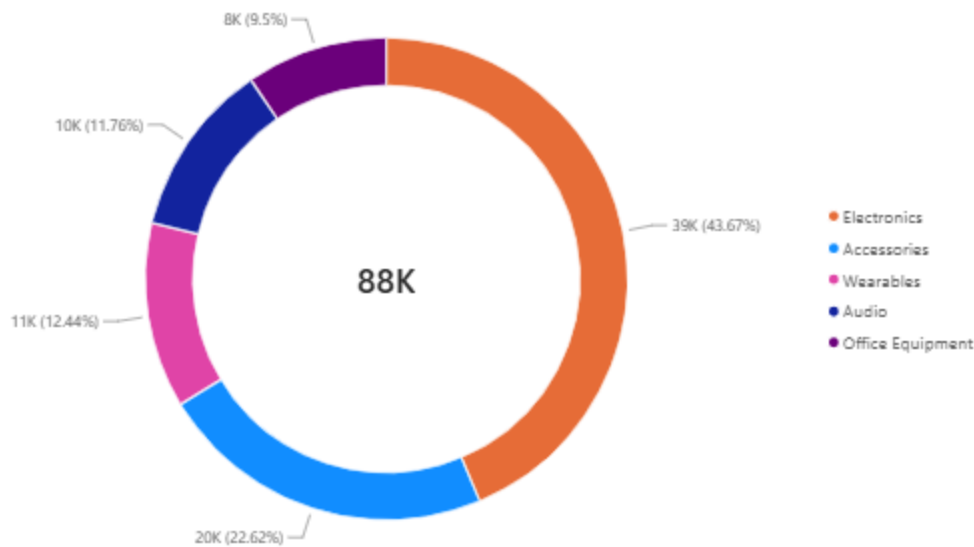


Sum of Sales Amount

by Year, Quarter, Month and Day



Sales Distribution by Category
Sum of Sales Amount by Category

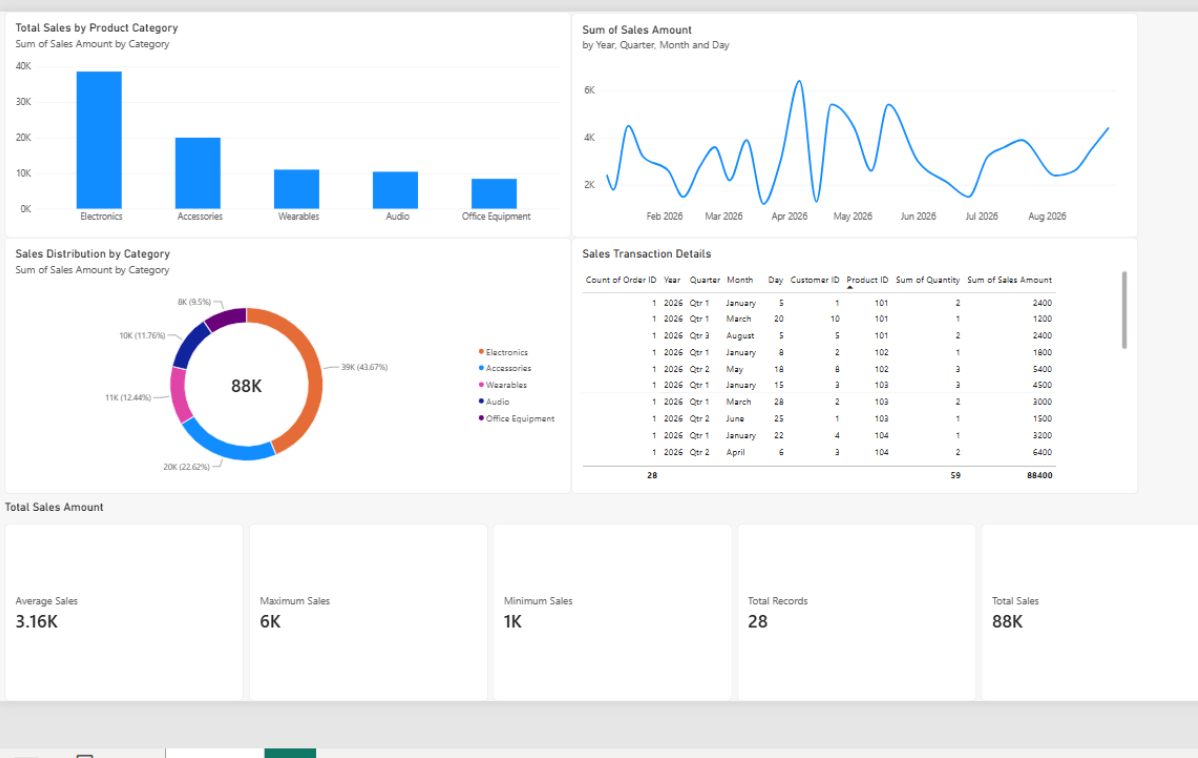


Sales Transaction Details

Count of Order ID	Year	Quarter	Month	Day	Customer ID	Product ID	Sum of Quantity	Sum of Sales Amount
1	2026	Qtr 1	January	5	1	101	2	2400
1	2026	Qtr 1	January	8	2	102	1	1800
1	2026	Qtr 1	January	15	3	103	3	4500
1	2026	Qtr 1	January	22	4	104	1	3200
1	2026	Qtr 1	February	3	5	105	2	2600
1	2026	Qtr 1	February	10	1	106	1	1500
1	2026	Qtr 1	February	18	6	107	4	2800
1	2026	Qtr 1	February	25	7	108	2	3600
1	2026	Qtr 1	March	4	8	109	1	2200
1	2026	Qtr 1	March	12	9	110	3	3900
1	2026	Qtr 1	March	20	10	101	1	1200
1	2026	Qtr 1	March	28	2	103	2	3000
1	2026	Qtr 2	April	6	3	104	2	6400
1	2026	Qtr 2	April	14	4	105	1	1300
1	2026	Qtr 2	April	21	5	108	3	5400
1	2026	Qtr 2	May	2	6	109	2	4400
1	2026	Qtr 2	May	10	7	110	2	2600
1	2026	Qtr 2	May	18	8	102	3	5400
1	2026	Qtr 2	June	1	9	106	2	3000
1	2026	Qtr 2	June	15	10	107	3	2100
28							59	88400

Total Sales Amount

Average Sales 3.16K	Maximum Sales 6K	Minimum Sales 1K	Total Records 28	Total Sales 88K
------------------------	---------------------	---------------------	---------------------	--------------------



Task 6: Interactive Dashboard & Filters

Create an interactive Power BI dashboard containing: At least 3 KPI Cards. At least 4 different visualizations. At least 2 slicers/filters. Appropriate titles and labels. Proper layout and organization. The dashboard should allow users to interact with and explore the data.

Interactive Dashboard

An interactive Power BI dashboard was created using the Sales dataset to provide a clear and user-friendly view of important sales information. The dashboard combines KPI cards, charts, tables, and slicers to allow users to analyze the data interactively.

KPI Cards

Three KPI Cards were included in the dashboard:

- **Total Sales** – Displays the total sales amount.
- **Average Sales** – Displays the average sales value.
- **Total Number of Records** – Displays the total number of sales records.

These KPI cards provide a quick summary of the overall sales performance.

Visualizations

The dashboard contains multiple visualizations created in Task 5:

1. **Total Sales by Product Category** – Column Chart used to compare sales between product categories.
2. **Sales Trend Over Time** – Line Chart used to analyze changes in sales over time.
3. **Sales Distribution by Category** – Donut Chart used to show the contribution of each category to total sales.

4. **Sales Transaction Details** – Table used to display detailed sales records.

Slicers/Filters

Two slicers were added to make the dashboard interactive:

- **Category Slicer** – Allows users to filter the dashboard according to product category.
- **Region Slicer** – Allows users to filter the dashboard according to customer region.

When a user selects a category or region, the relevant charts, KPI cards, and table update according to the selected filter.

Dashboard Layout and Organization

The KPI cards were placed at the top of the dashboard, followed by the charts and detailed table. The slicers were positioned on the side so that users can easily access the filters. Appropriate titles and labels were added to make each visualization easy to understand.

Interactivity

The dashboard allows users to select different categories and regions using the slicers. The visualizations and KPI cards respond to these selections, allowing users to explore the sales data from different perspectives.

Conclusion

An interactive Power BI dashboard was successfully created with three KPI Cards, four different visualizations, and two slicers. The dashboard provides an organized and interactive way to analyze sales performance and explore the dataset based on different categories and regions.

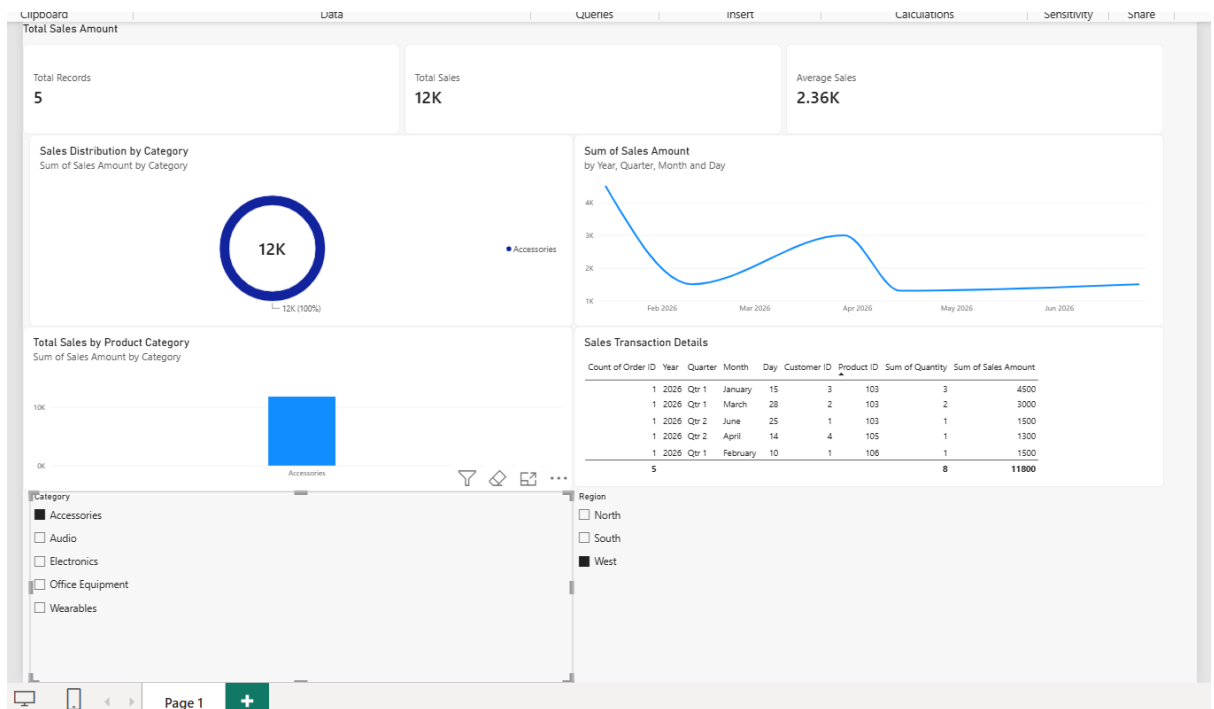
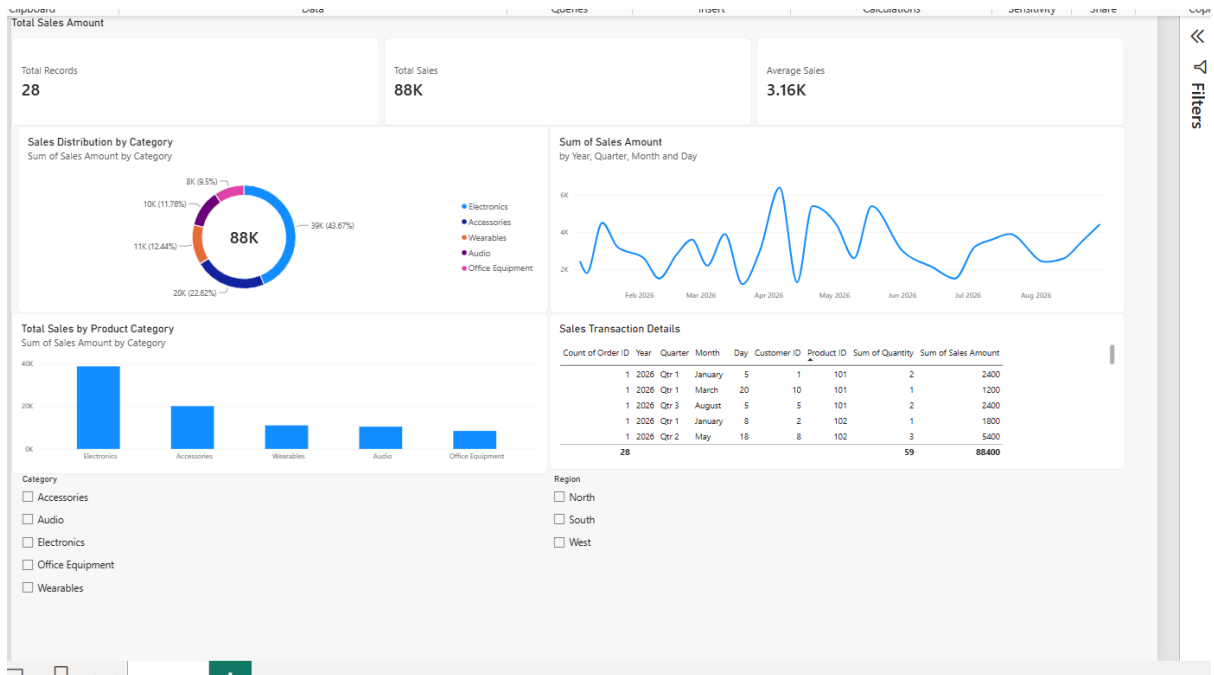
-----output-----

Category

- ☐ Accessories
- ☐ Audio
- ☐ Electronics
- ☐ Office Equipment
- ☐ Wearables

Region

- ☐ North
- ☐ South
- ☐ West



Task 7: Business Insights & Recommendations

Analyze your Power BI dashboard and write: At least 5 meaningful insights. Important trends or patterns identified from the dashboard. Highest and lowest performing categories where applicable. At least 2 business recommendations based on your findings. Your insights and recommendations should be supported by the data.

Business Insights

Based on the Power BI dashboard, the following insights were identified:

1. **Overall Sales Performance:**

The dashboard shows the overall sales performance based on the Total Sales KPI.

2. **Highest Performing Category:**

The category with the highest sales amount was **[Category Name]**, indicating stronger demand for products in this category.

3. **Lowest Performing Category:**

The category with the lowest sales amount was **[Category Name]**, indicating an area where sales performance can potentially be improved.

4. **Sales Trend:**

The sales trend over time shows **[increasing/decreasing/fluctuating]** sales, with the highest sales occurring around **[date/month]**.

5. **Regional Performance:**

The dashboard indicates that **[Region Name]** generated the highest sales, while **[Region Name]** had comparatively lower sales.

Business Recommendations

1. Focus on High-Performing Categories:

The business should maintain sufficient stock and consider increasing marketing efforts for the highest-performing categories to take advantage of customer demand.

2. Improve Low-Performing Categories/Regions:

The business should investigate the reasons for lower sales in underperforming categories or regions and consider targeted promotions, discounts, or marketing campaigns to improve their performance.

Conclusion

The Power BI dashboard helped identify important sales trends, category performance, and regional differences. These insights can be used by the business to make data-driven decisions and improve overall sales performance.

❖ Code link

https://drive.google.com/drive/folders/1suqCTtTR_OKbgtJqwRaV4bUaZd43DAXE?usp=sharing